Forensic Examinations



SecureWV

Tuan Phan

SecureWV 17 // Oct 22-23 // Charleston, WV

Into Insider Threats

About Tuan (Phanrensics)

AVP, eDiscovery and Digital Forensics.

EF/DF/DFIR, Data Security & Insider Threat Strategy



Senior Engineer

ThreatReel

Podcast

https://threatreel.com

Slide Deck & Contact

GitHub



https://github.com/phanrensics/slides

LinkedIn



https://www.linkedin.com/in/tuanqphan

Volunteer Work

Technical Mentor

CYBER ACADEMY

LAKOTA

Advisory Board

WARREN COUNTY CAREER CENTER

Cyber Reserve

OHIO CYBER RESERVE

OhCR

Disclaimer!

Yes, I have a day job. However…

Opinions expressed are based solely on my own independent security research and do not express or reflect the views or opinions of my employer.

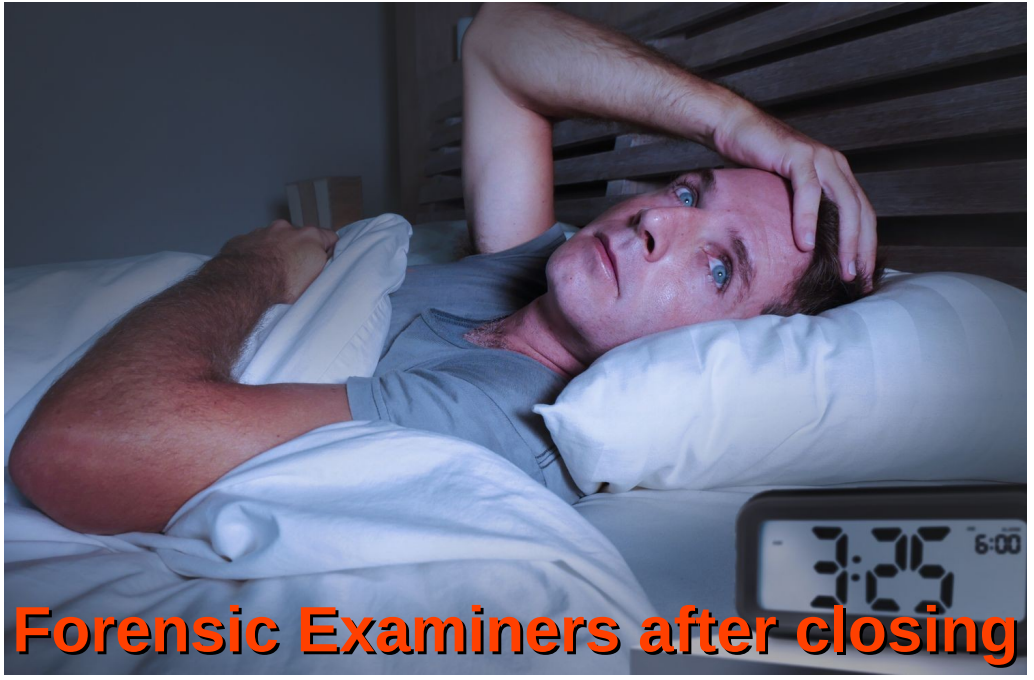


Agenda

Motivations, Risks, Stories

Detect & Investigate

- Lockout Bypasses
- Geolocation Spoofing
- Unauthorized Access



Insider Threat Motivations

- Unapproved Travel
- Over-employment Schemes
- Unauthorized Contracting
- Nation-state Agents & Operatives (Mules)

Organizational Risks

- Financial Loss
- Regulatory Penalties
- Operational Downtime
- Erosion of Trust

True Stories

- Wells Fargo
- DPRK Fraudulent IT Workers

True Story: Wells Fargo

Dozen+ Employees Fired

- Location?



Simulating Keyboard Activity

https://www.cbsnews.com/news/wells-fargo-fires-employees-faking-work/

True Stories: DPRK

DPRK (North Korean)

- Agents
- Mules



Fraudulent IT Workers

- Stolen Identities

https://www.justice.gov/opa/pr/justice-department-announces-coordinated-nationwide-actions-combat-north-korean-remote

Account Lockout Bypasses

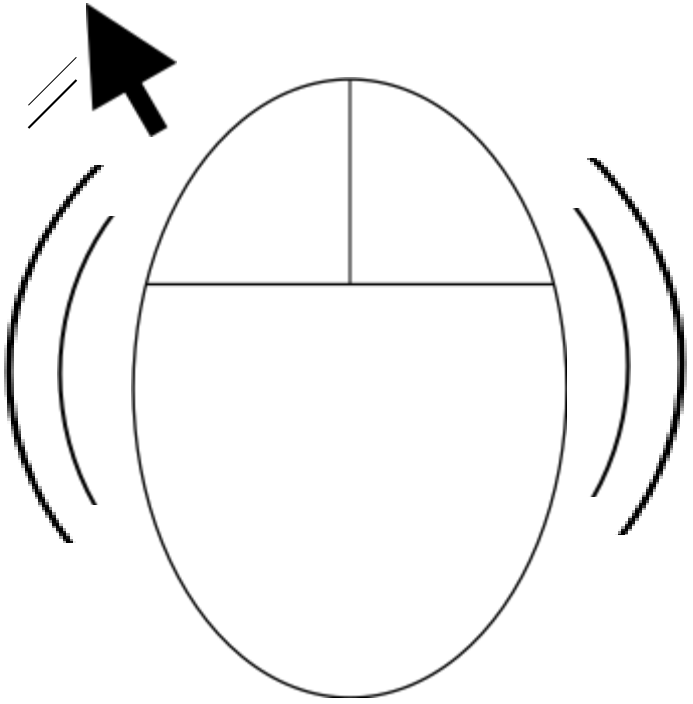
- Techniques
  - Mouse Jigglers
  - Keyboard Key Togglers

Additional Risks

- Policy Violations
- Unattended & Unlocked Systems
- Stolen Productivity & Integrity
- Potentially Hides Trojan'ed Malware

Mouse Jigglers

- Hardware & Software
- Also called a
  - Mouse Mover
  - Mouse Buddy

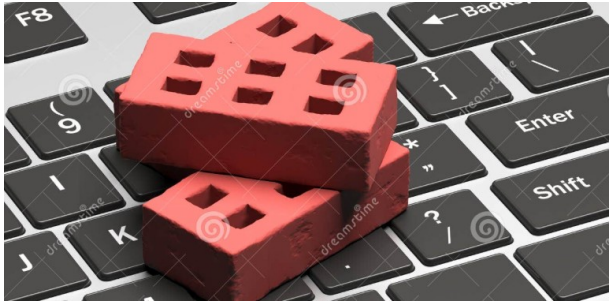


Hardware Devices

- Online Stores
  - USB dongles (HID)
  - Mechanical Platforms





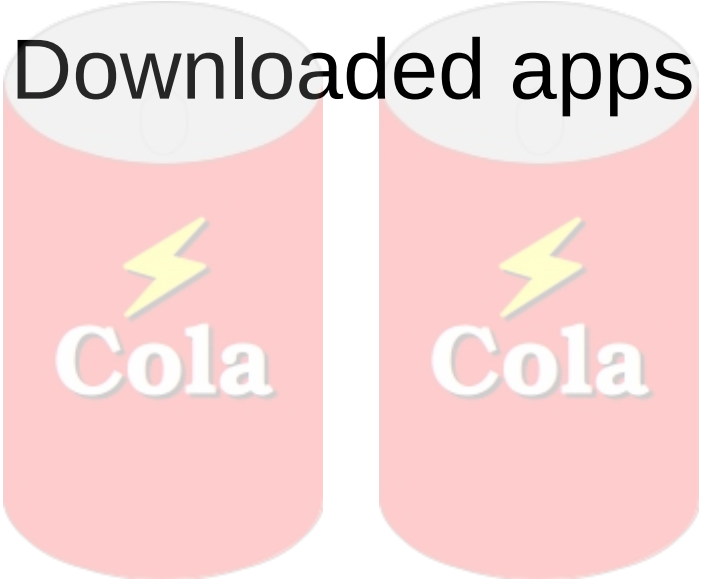






Software Jigglers & Togglers

- caffeine.exe
- Downloaded apps

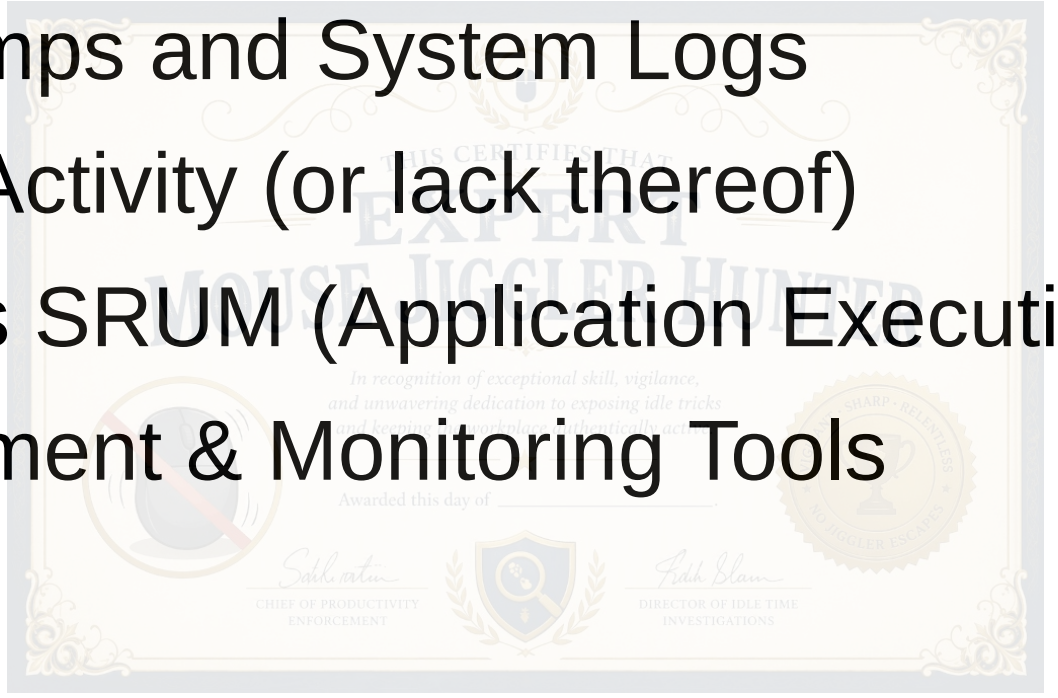


- Custom scripts, macros, etc.

```
min = 5
loop {
    sleep, % min*60*1000
    mousemove, 10, 0, 20, R
    mousemove, -10, 0, 20, R
}
```

Behavioral Detections

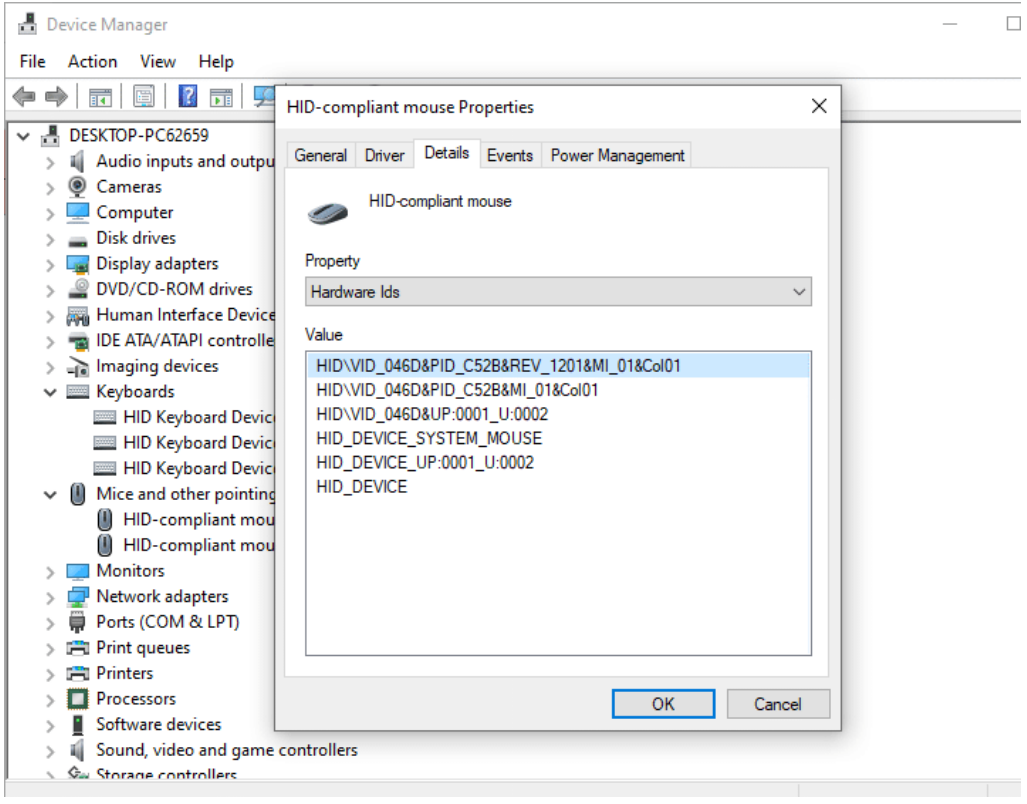
- Timestamps and System Logs
- System Activity (or lack thereof)
- Windows SRUM (Application Execution)
- Management & Monitoring Tools



Device Signatures

- Vendor ID (VID): 0x0E90
- Product ID (PID): 0x0028 or 0x0045
- USB device class
  - Windows Registry, USBSTOR, syslog

Windows Artifact Example



Further Reading

"How to Detect Mouse Jiggler Activity in the Workplace”

By: Jahnavi Pulluri



https://www.timechamp.io/blogs/how-to-detect-mouse-jiggler-activity-in-the-workplace/

Geolocation Spoofing



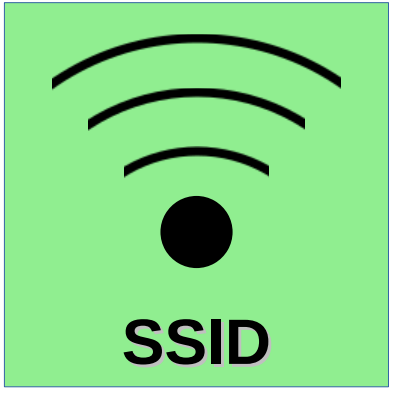
Laptop Farms

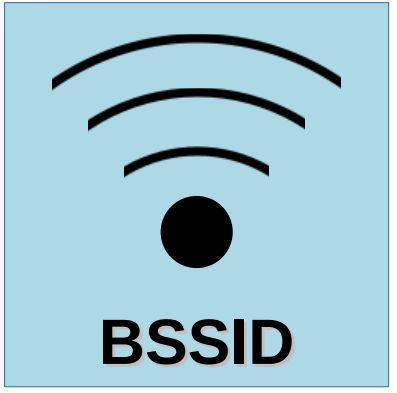


Connectivity Artifacts

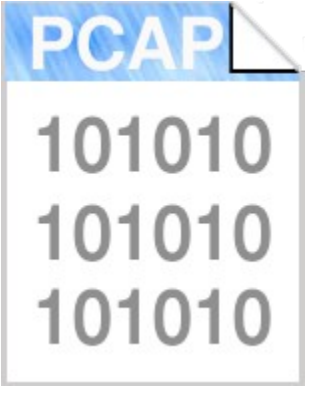










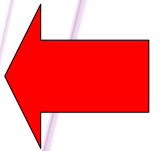


VPN Logs

log_timestamp	user_dst	city_sour	state_sour	country_sc	postal_so	longitude_so	latitude_sour	event_description	ip_src
2026-01-01T17:38:09.000-06:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	passed_hc_policies:['OS_Check$$ Asset Certificate']
2026-01-14T13:30:38.000-06:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	passed_hc_policies:[' Asset Certificate$$OS_Check']
2026-03-02T19:32:44.000-06:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	75.66.207.3
2026-03-10T17:29:30.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	75.66.207.3
2026-04-09T10:28:47.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-04-12T11:33:02.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-04-26T21:54:26.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Closed connection	75.66.207.3
2026-04-27T10:23:47.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-04-29T10:31:31.000-05:00	JDOE999	Memphis	Tennessee	United States	38115	-89.8538	35.056	Agent login succeeded	primary auth-server name: [2factor_auth]
2026-05-26T15:04:19.000-05:00	JDOE999	Memphis	Tennessee	United States	38122	-89.9274	35.1567	Closed connection	75.66.207.3

Proximity & Proxy Logs

- Localized Web Traffic
  - Example: google.com vs **google.co.??**



- Hotel Wi-Fi or other “Captive Portal” locations

PROXY

Wi-Fi Profile

```
netsh wlan show profile
```

```
Microsoft Windows [Version 10.0.18363.836]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\om>netsh wlan show profile

Profiles on interface Wi-Fi:

Group policy profiles (read only)
---------------------------------
    <None>

User profiles
-------------
    All User Profile     : Moto G Play 1087 2
    All User Profile     : vivo Y31L
    All User Profile     : vivo 1802
    All User Profile     : FLGM-bWVnaGFrYWtrYXI5MDU4
    All User Profile     : Moto G Play 1087
    All User Profile     : LPU Hostels
    All User Profile     : JioNet@LPU
    All User Profile     : LPU Wireless
    All User Profile     : Den Boomband


C:\Users\om>
```

Wi-Fi Interface

```
netsh wlan show interface
```

```
There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : Intel(R) Dual Band Wireless-AC 3165
    GUID                   : d0f2b8e0-66a5-4b6d-bc3c-e3b30baa95fd
    Physical address       : 00:e1:8c:31:4d:cc
    State                  : connected
    SSID                   : My-Network
    BSSID                  : fc:dd:55:67:bb:b9
    Network type           : Infrastructure
    Radio type             : 802.11n
    Authentication         : WPA2-Personal
    Cipher                 : CCMP
    Connection mode        : Profile
    Channel                : 5
    Receive rate (Mbps)    : 150
    Transmit rate (Mbps)   : 150
    Signal                 : 99%
    Profile                : My-Network

    Hosted network status  : Not available
```

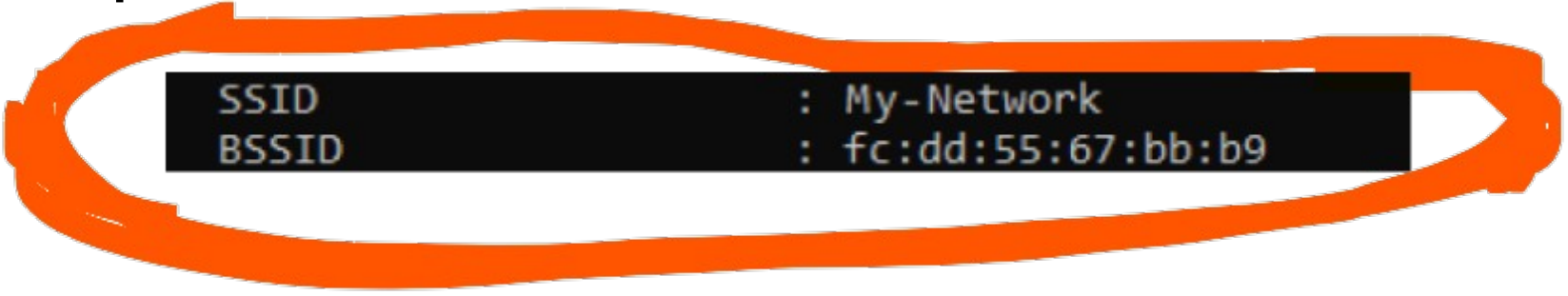
SSID & BSSID

Service Set Identifier (SSID)

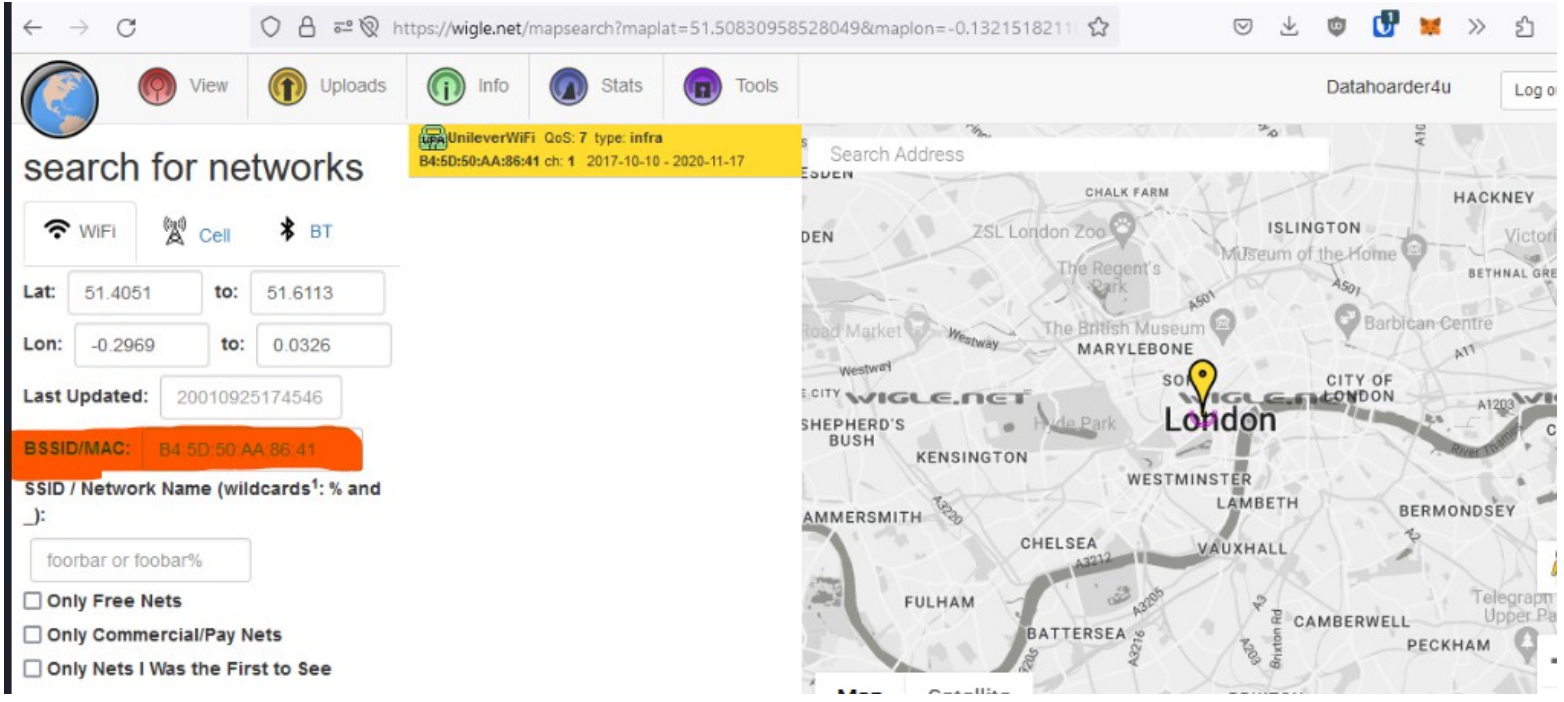
- Wireless Access Point (AP) Name

Basic Service Set Identifier (BSSID)

- Unique AP MAC Address

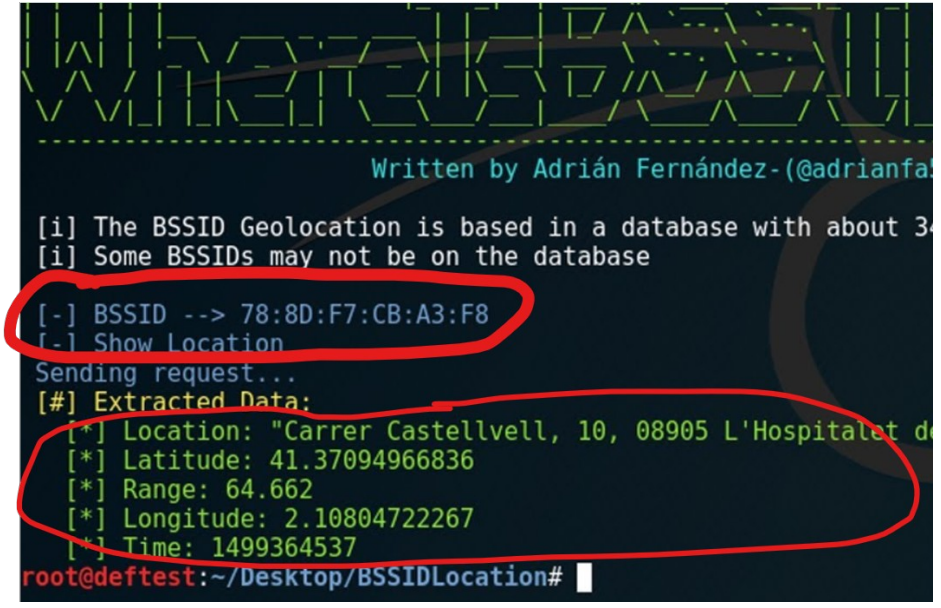


“wigle.net” Lookup



“whereisbssid.py” Lookup

https://github.com/Trackbool/WhereIsBSSID/

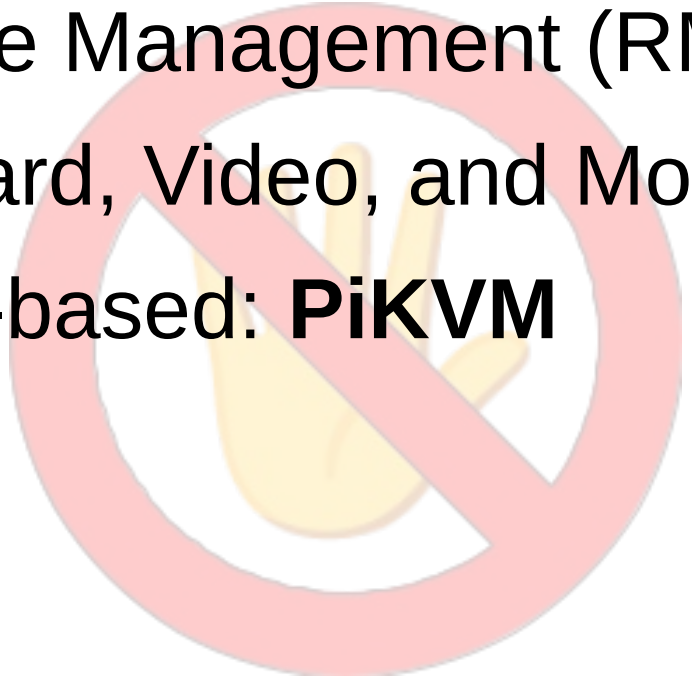


Unauthorized Access

Remote Machine Management (RMM) Apps

Network Keyboard, Video, and Mouse (IP-KVM)

- Raspberry Pi-based: **PiKVM**



Hunting RMM Apps

“RMM Threat Intelligence Tracker”

By: Threat-Hunting-Journal



https://github.com/Threat-Hunting-Journal/RMM-Tool-Tracking

The RMM that’s Built-In

“We Need to Talk About Microsoft Quick Assist”

By: Tyler Hudak

https://inversion6.com/insights/blog/microsoft-quick-assist-an-it-security-primer

PiKVM

- Network connected & USB
- 1 device, multiple hosts
- ~$300 USD



PiKVM Security Risks

- Internet Accessibility
- Works like any other USB peripheral
- Easily abused for unauthorized access

Network Detection

DNS, Proxy, and SSH

- Connection strings with “pikvm” as a username
- Potential mismatches of SSH host keys in the “**known_hosts**” file

Other Strategies

EDR telemetry

MAC address vendor OUI matches

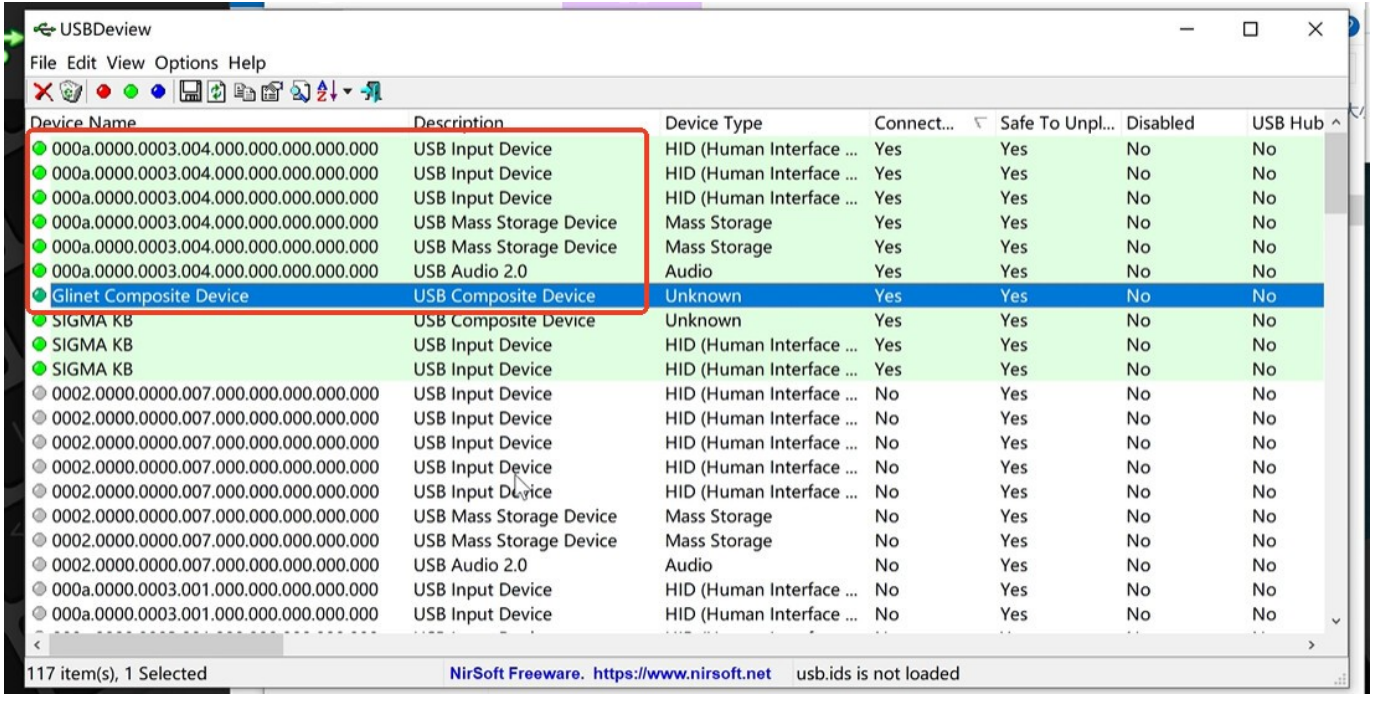
- Sispeed, Raspberry Pi Foundation, GL.iNET

USB Detections

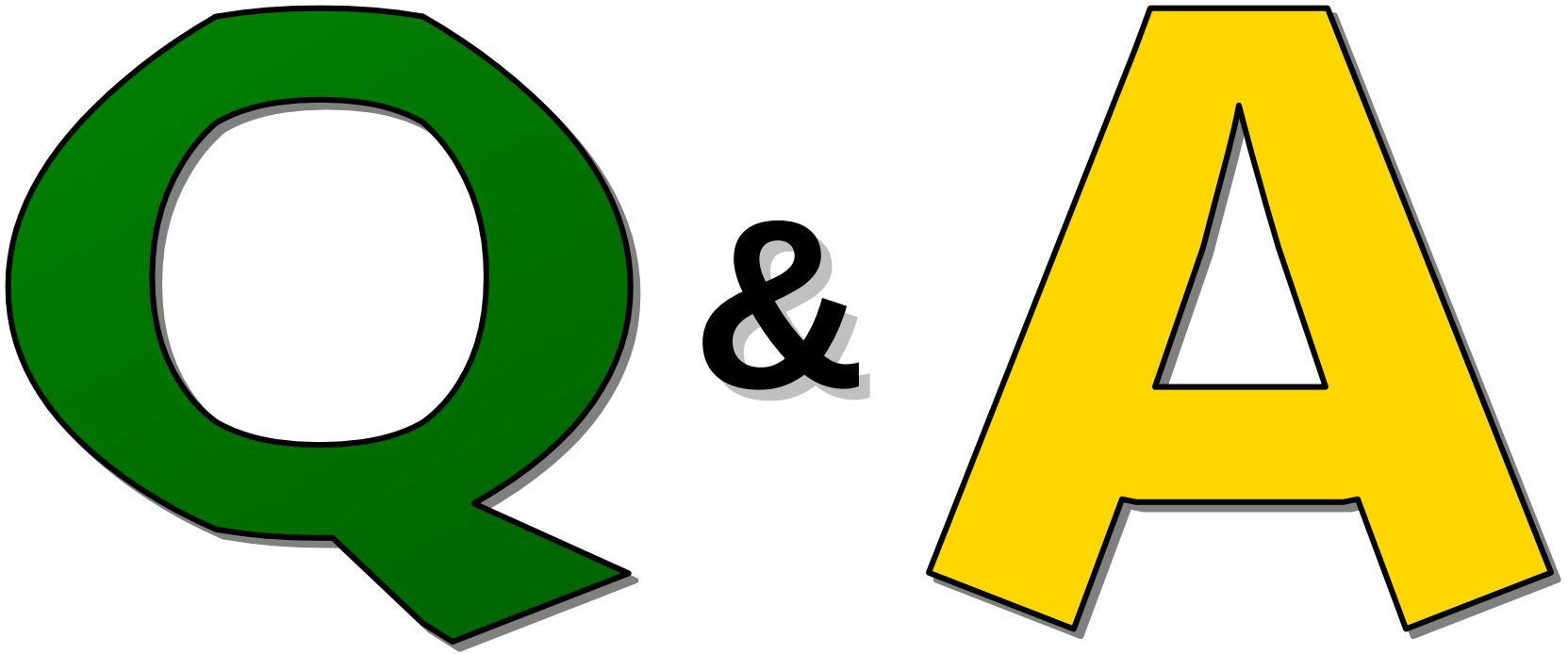
- Hardware IDs or descriptors
  - Particularly generic USB device names (KVM?)
- **lsusb** (Linux)
- Device Manager (Windows)



USB Example


USBDeview
File Edit View Options Help
Device Name | Description | Device Type | Connect... | Safe To Unpl... | Disabled | USB Hub
000a.0000.0003.004.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | Yes | Yes | No | No
000a.0000.0003.004.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | Yes | Yes | No | No
000a.0000.0003.004.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | Yes | Yes | No | No
000a.0000.0003.004.000.000.000.000.000 | USB Mass Storage Device | Mass Storage | Yes | Yes | No | No
000a.0000.0003.004.000.000.000.000.000 | USB Mass Storage Device | Mass Storage | Yes | Yes | No | No
000a.0000.0003.004.000.000.000.000.000 | USB Audio 2.0 | Audio | Yes | Yes | No | No
Glinet Composite Device | USB Composite Device | Unknown | Yes | Yes | No | No
SIGMA KB | USB Composite Device | Unknown | Yes | Yes | No | No
SIGMA KB | USB Input Device | HID (Human Interface ... | Yes | Yes | No | No
SIGMA KB | USB Input Device | HID (Human Interface ... | Yes | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Mass Storage Device | Mass Storage | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Mass Storage Device | Mass Storage | No | Yes | No | No
0002.0000.0000.007.000.000.000.000.000 | USB Audio 2.0 | Audio | No | Yes | No | No
000a.0000.0003.001.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
000a.0000.0003.001.000.000.000.000.000 | USB Input Device | HID (Human Interface ... | No | Yes | No | No
117 item(s), 1 Selected
NirSoft Freeware. https://www.nirsoft.net
usb.ids is not loaded

Questions?



The End



Thank you for Staying After Closing with **Forensic Examinations into Insider Threats!**